Final Year Project

FYP Group

Muhammad Hussnain Kasana 22F-3139

Muhammad Shahbaz Sarwar 22F-3151

Muhammad Shoaib Haider 22F-3113

Supervised by

Dr. Muhammad Fayyaz

Department of Artificial Intelligence

National University of Computer and Emerging Science Chiniot-Faisalabad Campus, Pakistan 2025

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Abstract

The healthcare industry generates vast amounts of billing and insurance data daily, yet much of the process remains manual, error-prone, and time-consuming. HealPay aims to automate the medical billing lifecycle using artificial intelligence to enhance accuracy, speed, and transparency. The system integrates Optical Character Recognition (OCR) and Natural Language Processing (NLP) to extract and interpret medical data from patient records, maps diagnoses and procedures to standard ICD and CPT codes, and uses machine learning to detect billing anomalies and fraud patterns. HealPay verifies insurance eligibility, generates invoices automatically, and provides real-time claim tracking for both healthcare providers and patients. This system improves revenue cycle management, supports compliance with medical coding standards, and enables faster, more reliable reimbursements across the U.S. medical billing ecosystem .

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1. Introduction

Healpay contains 5 entities Patient, Doctor, Medical Coder, Admin and Billing Staff. The system will provide services for automating the process of billing helping the doctor to process the patient information easily, recommending codes to the medical coder, verify insurance claims and detect billing anomalies while admin will be managing all these entities.

2. Vision Document

In this section, project vision is discussed in detail.

2.1. Problem Statement

Table 1: Problem Statement

Problem of	1. Manual data entry and Coding 2. Leading to frequent errors and inefficiencies
Affects	1. Hospitals and Insurance Companies face delays in claim processing 2. Medical Coder and Billing Staff experience high workloads and repetitive task
Impact of which	1. Increased claim rejections and revenue losses 2. Administrative costs rise due to repeated manual corrections
Successful solution would be	1. AI-driven Solution that automates billing and coding 2. Automated anomaly detection to prevent errors 3. Reducing manual effort and improving accuracy

2.2. Business Opportunity

The healthcare billing sector faces growing demand for automation and cost reduction. HealPay offers a scalable AI-driven solution that cuts manual effort, minimizes errors, and accelerates claim processing. Its anomaly detection and integration-ready design make it valuable for

hospitals, clinics, and billing agencies seeking reliable, data-driven financial management.

2.3. Objectives

- To automate the end-to-end medical billing process using Artificial Intelligence.
- To extract and process patient and diagnosis data through OCR and NLP.
- To map medical data accurately to standard ICD and CPT codes.
- To verify insurance eligibility and streamline claim submission.
- To detect billing anomalies and potential fraud using machine learning models.
- To provide a dashboard for tracking claims, invoices, and payment status.
- To reduce manual effort, errors, and administrative time in healthcare billing.

2.4. Scope

The scope of HealPay encompasses the automation of the complete medical billing workflow, from data extraction to payment tracking. The system is designed to handle key operations such as patient data processing through OCR and NLP, automated mapping of diagnoses and procedures to ICD and CPT codes, claim generation, insurance verification, and invoice management. It supports multiple user roles including Doctor, Medical Coder, Billing Staff, Patient, and Administrator, each with dedicated interfaces and functionalities. The system also integrates AI-driven anomaly detection to identify potential billing errors or fraud before claim submission. While primarily aligned with U.S. healthcare billing standards, HealPay is scalable for adaptation across different healthcare environments, ensuring faster, more accurate, and transparent billing operations.

2.5. Constraints

- Requires high-quality medical documents for accurate OCR extraction.
- Dependent on reliable internet connectivity for cloud-based AI operations.
- Integration with real hospital databases and insurance APIs may be limited in prototype phase.
- AI model accuracy depends on the size and quality of the training dataset.
- Compliance with medical data privacy regulations (e.g., HIPAA) must be ensured in real deployments.

2.6. Stakeholder and User Descriptions

Stakeholder and User Descriptions are defined in detail below.

2.6.1. Market Demographics

The target market for HealPay primarily includes healthcare institutions and stakeholders involved in the medical billing ecosystem. This consists of hospitals, private clinics, diagnostic centers, medical billing companies, and insurance providers operating within the United States. These organizations face rising administrative costs, claim processing delays, and compliance challenges under the existing manual systems.

The secondary market involves third-party billing service providers and healthcare startups seeking cost-effective, AI-enabled automation tools. With the growing emphasis on digital transformation in healthcare and the increasing adoption of AI-driven solutions, HealPay fits into a rapidly expanding sector. The U.S. medical billing and coding market alone is projected to grow significantly in the coming years, offering strong potential for large-scale adoption and commercial viability.

2.6.2. Stakeholder Summary

Table 2: Stakeholder Summary

Name	Description	Responsibilities
HealPay-Developers	It includes those who are involved in designing and implementing of this project.	1. Should design a website considering user's needs 2. Proper testing should be applied
End-User	End users include all doctor, patient, medical coder, billing staff and admin.	1. Should have knowledge of using web interface application
Supervisors of project	Supervisor will be stakeholders of this product as they are involved in development process of this project with the development team.	1. Gives direction to development team 2. Ensures that the system will be maintainable 3. Ensures that there will be a market demand for the product's features 4. Monitors the project's progress 5. Ensures that project is in compliance with the documents generated during project planning and work products are properly delivered

2.6.3. User Environment

HealPay is designed as a web-based application accessible through standard browsers on desktops and laptops. It provides secure login-based access for different user roles, including doctors, medical coders, billing staff, patients, and administrators. Each user interacts with a tailored dashboard suited to their responsibilities, such as data entry, claim verification, or billing review. The system requires an internet connection for cloud-based processing and supports a user-friendly interface that ensures ease of navigation even for non-technical users.

2.6.4. Stakeholder Profiles

The Stakeholder Profiles Attributes of the system are listed in this section. *2.6.4.1. Supervisors of the Project*

Table 3: Supervisors of the Project

Representatives	Supervisor: Dr. Muhammad Fayyaz
Description	They are involved in supervising activities of development process
Type	They are technical stakeholders. They have expertise in domains which are being applied to this project i.e. machine learning, web development.
Responsibilities	1. Gives direction to development team 2. Ensures that the system will be maintainable 3. Ensures that there will be a market demand for the product's features 4. Monitors the project's progress 5. Ensures that project is in compliance with the documents generated during project planning and work products are properly delivered 6. They will facilitate development team to complete this project within specified resources.
Success Criteria	The completion of features which are being committed by development team at start of the project.

Involvement	1. Requirement reviewer 2. Senior managers 3. Reviews implementation
Comments/Issues	None

2.6.4.2. Development Team of the Project

Table 4: Development Team of the Project

Representatives	Mr. Muhammad Shoaib Haider Mr. Muhammad Hussnain Kasana Mr. Muhammad Shahbaz Sarwar
Description	They are involved in designing of web application.
Type	They are technical stakeholders.
Responsibilities	1. Should design this web application considering user's needs 2. Proper testing should be applied to make it as effective as possible 3. Proper data is required for better recommendations
Success Criteria	The completion of features which are being committed by development team at start of the project.
Involvement	1. Designers 2. Testers
Deliverable	1. Documentation 2. Data acquisition 3. System Training
Comments/Issues	None

3. System Requirement Specification

In this section, features and requirements of the system are explained.

3.1. System Features

- User Authentication & Role Management
- Patient Data Entry
- OCR-Based Data Extraction
- Code Recommendation Engine
- ICD & CPT Code Mapping
- Automated Claim Form Generation(CMS-1500)
- Anomaly & AI Claim Prediction
- Reporting Dashboard
- Notifications & Alerts

3.2. Functional Requirements

The functional requirements of the system are listed below.

3.2.1. Patient Registration and Intake

The system must allow patients to securely register or log in. Patients need to be able to fill out their personal details and insurance information digitally, potentially using scanned documents. Once logged in, they should have access to a dashboard to view their billing statements, check the status of their insurance claims, make payments, and receive notifications.

3.2.2. Clinical Encounter Documentation

Doctors/Clinicians must be able to document patient encounters within the system or have the system integrate with an existing EHR. This involves recording diagnoses, procedures performed, and any relevant clinical notes. This information forms the basis for medical coding and billing.

3.2.3. Medical Coding Assistance

Medical coders must be able to access the documented encounter information. The system should assist them by processing medical notes (using NLP) and suggesting relevant ICD-10 and CPT codes

based on AI analysis. Coders need tools to review these suggestions, manually assign or correct codes, and validate the final code assignments against standard rules.

3.2.4. Claim Generation and Validation

The system must automatically generate standard insurance claim forms (like CMS-1500) based on the coded encounter data. Before submission, the system needs to perform automated validation ("scrubbing") to check for common errors, missing information, or code mismatches. It should also use AI to predict the likelihood of the claim being rejected by the payer.

3.2.5. Claim Submission and Tracking

Billing staff must be able to submit the validated electronic claims to insurance payers through integrated clearinghouses. The system needs to track the status of each submitted claim (e.g., accepted, rejected, paid) by receiving updates from the payer. This status should be visible to both billing staff and patients (for their claims).

3.2.6. Rejection Management

When a claim is rejected or denied by the payer, the system must capture the reason code and details. It should provide tools and workflows for billing staff to review the rejection, make necessary corrections (like adding documentation or fixing codes), and manage the appeal or resubmission process.

3.2.7. Payment Processing

The system must allow patients to make payments online through a secure portal. It should also enable billing staff to manually post payments received (e.g., checks, insurance payments/remittance advice) and apply them to the correct patient accounts and claims.

3.2.8. System Administration and Configuration

System administrators need tools to manage user accounts, roles, and permissions. They must also be able to configure system settings, maintain master data like code databases, and monitor the performance of the system and its AI components.

3.3. Non-Functional Requirements

The functional requirements of the system are listed below.

3.3.1. Security Requirements (HIPAA Compliance)

Keeping patient information safe and private is a top priority. HealPay must follow the strict US health privacy law called HIPAA .This means all sensitive data needs to be scrambled (encrypted) when stored or sent over the internet. Only authorized users should be able to see specific information based on their role (like a patient only seeing their own bills), and the system must keep a log of who accessed what data. Users also need secure ways to log in.

3.3.2. Performance Requirements

HealPay needs to be fast and responsive. Users shouldn't have to wait long for pages to load, especially dashboards showing financial information. Critical tasks like checking insurance eligibility or getting AI suggestions for codes should happen quickly (within seconds). The system should handle many users and process claims efficiently without slowing down.

3.3.3. Usability Requirements

The system must be easy to learn and use, even for people who aren't tech-savvy. The patient portal, in particular, needs to be super clear, avoiding confusing medical or insurance jargon, so patients can easily understand their bills and what they owe. For staff (coders, billers), the system should make their workflows efficient, helping them find information and complete tasks with minimal clicks.

3.3.4. Reliability Requirements

HealPay must be dependable and available when needed. Since it handles important financial tasks for clinics, it should rarely crash or be offline unexpectedly (aiming for 99.9% uptime). The system

must also protect against data loss or corruption, especially during critical operations like payment processing or claim submission.

3.3.5. Compliance Requirements

Besides HIPAA, the system must work correctly with US healthcare standards. This includes using the right formats for medical codes (like ICD-10, CPT) and being able to exchange data electronically with insurance companies using standard formats (like EDI or APIs like FHIR). This ensures HealPay can actually connect and communicate properly within the US healthcare system.

3.4. Other Requirements

Some other requirements of the system are listed below.

3.4.1. Application Standards

HealPay will be a web-based application. This means users (patients, staff, admins) will access it using a standard web browser like Chrome, Firefox, Safari, or Edge on their computers or mobile devices. It doesn't require installing special software on each user's machine

3.4.2. Performance Requirements

- The web application would support one user against the database on server at any given time.
- The web application would access the server Database within no time.
- The web application would compute results effectively and quickly.

3.4.3. Document Requirements

The User Manual would describe use of the website from patient, doctor, medical coder and admin’s view point. It will be available within the application. The User Manual should include:

- Minimum System Requirements
- All System Features

4. Use Case Diagram

This Use Case Diagram models the HealPay Medical Billing System's functional requirements, depicting six primary actors (Patient, Doctor, Medical Coder, Billing Staff, Insurance Payer, and System Administrator) and their interactions across the complete billing lifecycle from patient registration through payment processing. The diagram illustrates core processes including encounter documentation, medical coding, claim generation, validation, submission, and payment handling, while also showing mandatory authentication (via include relationships) and optional AI-powered enhancements for code recommendations and rejection prediction (via extend relationships).

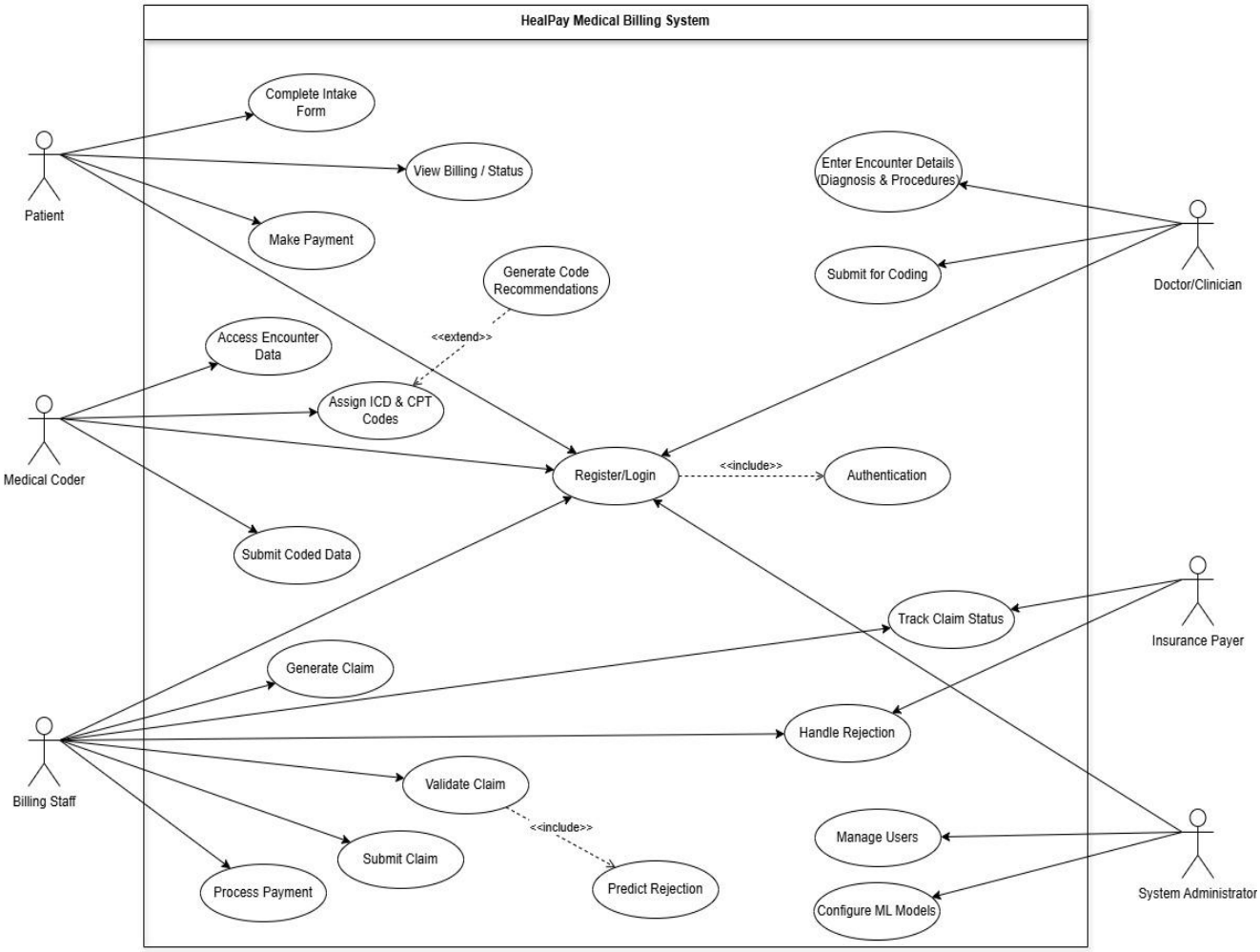

Figure 1: Use case Diagram for HealPay

5. High Level Use case

5.1. Login

Table 5: UC-01 Login

Use case ID	01
Use case Name	Login
Actor	Doctor, Patient, Medical Coder, Billing Staff, Admin
Type	Primary
Description	The user would have to login to the system to continue using the application

5.2. Authentication

Table 6:UC-02 Authentication

Use case ID	02
Use case Name	Authentication
Actor	System (Internal Process)
Type	Include (Included by Register/Login)
Description	System verifies user credentials against stored records and establishes secure session for authorized access.

5.3. Complete Intake Form

Table 7: UC-03 Complete Intake Form

Use case ID	03
Use case Name	Complete Intake Form
Actor	Patient
Type	Primary
Description	The patient provides their personal, demographic, and insurance information to the HealPay system before or during their visit.

5.4. Enter Encounter Details

Table 8: UC-04 Enter Encounter details

Use case ID	04
Use case Name	Enter Encounter details
Actor	Doctor/Clinician
Type	Primary
Description	Doctor documents patient diagnosis, treatment procedures, medications, and clinical notes during or after patient visit.

5.5. Submit For Coding

Table 9: UC-05 Submit For Coding

Use case ID	05
Use case Name	Submit For Coding
Actor	Doctor/Clinician
Type	Primary
Description	Doctor finalizes encounter documentation and submits it to medical coding department for ICD and CPT code assignment.

5.6. Access Encounter Data

Table 10: UC-06 Access Encounter Data

Use case ID	06
Use case Name	Access Encounter Data
Actor	Medical Coder
Type	Primary
Description	Medical coder retrieves and reviews encounter documentation submitted by doctors to prepare for code assignment.

5.7. Assign ICD & CPT Codes

Table 11: UC-07 Assign ICD & CPT Codes

Use case ID	07
Use case Name	Assign ICD & CPT Codes
Actor	Medical Coder
Type	Primary
Description	The coder reviews clinical documentation and assigns the appropriate standardized medical codes (ICD-10, CPT) for billing purposes, potentially aided by system recommendations

5.8. Generate Code Recommendations

Table 12: UC-08 Generate Code Recommendations

Use case ID	08
Use case Name	Generate Code Recommendations
Actor	System (ML/AI Module)
Type	Extend (Extends Assign ICD & CPT Codes)
Description	AI system analyzes encounter documentation and suggests relevant ICD and CPT codes to assist medical coder.

5.9. Submit Coded Data

Table 13: UC-09 Submit Coded Data

Use case ID	09
Use case Name	Submit Coded Data
Actor	Medical Coder
Type	Primary
Description	Medical coder finalizes code assignment and submits coded encounter to billing department for claim generation.

5.10. Generate Claim

Table 14: UC-10 Generate Claim

Use case ID	10
Use case Name	Generate Claim
Actor	Billing Staff
Type	Primary
Description	Billing staff creates insurance claim using coded encounter data, patient information, and insurance details.

5.11. Validate Claim

Table 15: UC-11 Validate Claim

Use case ID	11
Use case Name	Validate Claim
Actor	Billing Staff
Type	Primary
Description	Billing staff reviews generated claim for accuracy, completeness, and compliance with payer requirements before submission.

5.12. Predict Rejection

Table 16: UC-12 Predict Rejection

Use case ID	12
Use case Name	Predict Rejection
Actor	System (ML/AI Module)
Type	Extend (Extends Validate Claim)
Description	AI system analyzes claim data against historical rejection patterns and predicts likelihood of claim rejection.

5.13. Submit Claim

Table 17: UC-13 Submit Claim

Use case ID	13
Use case Name	Submit Claim
Actor	Billing Staff
Type	Primary
Description	Billing staff electronically submits validated claim to insurance payer through clearinghouse or direct submission.

5.14. Track Claim Status

Table 18: UC-14 Track Claim Status

Use case ID	14
Use case Name	Track Claim Status
Actor	Insurance Payer, Billing Staff
Type	Primary
Description	System monitors claim status with insurance payer and updates billing staff on acceptance, rejection, or payment processing.

5.15. View Billing Status

Table 19: UC-15 View Billing Status

Use case ID	15
Use case Name	View Billing Status
Actor	Patient
Type	Primary
Description	Patient accesses their billing information to view charges, insurance payments, outstanding balance, and claim status.

5.16. Manage Users

Table 20: UC-15 Manage Users

Use case ID	16
Use case Name	Manage Users
Actor	System Administrator
Type	Primary
Description	Administrator creates, modifies, deactivates user accounts and assigns appropriate roles and permissions within the system.

5.17. Configure ML Models

Table 21: UC-17 Configure ML Models

Use case ID	17
Use case Name	Configure ML Models
Actor	System Administrator
Type	Primary
Description	Administrator configures AI/ML model parameters, retrains models with new data, and monitors model performance metrics.

6. Expended Use case

6.1. Login

Table 22: EUC-01 Login

Use case ID	01
Use case Name	Login
Actor	Patient, Doctor/Clinician, Medical Coder, Billing Staff, System Administrator
Description	Users create accounts or authenticate themselves to access the HealPay system with role-based permissions.
Trigger	User navigates to the system login page or clicks register button
Pre-Condition	System is accessible and database is operational
Post Condition	User is authenticated and redirected to their role-specific dashboard
Normal Flow	• User enters credentials •System validates credentials •Authentication is performed •User gains access to dashboard
Alternative Flow	•Invalid credentials entered - display error message •New user registration - collect user details and create account •Forgot password - initiate password reset flow
Special Requirement	Availability of Internet
Frequency of Use	High
Assumption	User has valid credentials or registration information

6.2. Authentication

Table 23: EUC-02 Authentication

Use case ID	02
Use case Name	Authentication
Actor	System (Internal Process)
Description	System verifies user credentials against stored records and establishes secure session for authorized access.
Trigger	Register/Login use case is initiated
Pre-Condition	User credentials are submitted, database contains user records
Post Condition	Authentication token is generated, user session is established
Normal Flow	• Receive credentials • Query database for user record • Validate password hash • Generate session token • Return authentication status
Alternative Flow	• Credentials don't match - return authentication failure • Account locked - notify user and deny access
Special Requirement	SSL/TLS encryption, secure token generation, password hashing algorithm
Frequency of Use	High
Assumption	Database is accessible and authentication service is running

6.3. Complete Intake Form

Table 24: EUC-03 Complete Intake Form

Use case ID	03
Use case Name	Complete Intake Form
Actor	Patient
Description	Patient fills out medical history, insurance details, and personal information required for billing and treatment documentation.
Trigger	Patient initiates new visit or appointment process
Pre-Condition	Patient is logged into the system
Post Condition	Intake form data is saved and available for doctor and billing staff
Normal Flow	• Patient accesses intake form • Fills in personal details, medical history, and insurance information • Validates required fields • Submits form • System stores data
Alternative Flow	• Incomplete form - highlight missing fields and prompt completion • Invalid insurance information - request verification • Save as draft - allow patient to complete later
Special Requirement	HIPAA compliance, data encryption, form validation
Frequency of Use	Medium
Assumption	Patient has necessary information available (insurance card, medical history)

6.4. Enter Encounter Details

Table 25: EUC-04 Enter Encounter Details

Use case ID	04
Use case Name	Enter Encounter Details
Actor	Doctor/Clinician
Description	Doctor documents patient diagnosis, treatment procedures, medications, and clinical notes during or after patient visit.
Trigger	Doctor completes patient examination
Pre-Condition	Patient intake form is completed, doctor is authenticated
Post Condition	Encounter details are saved and ready for coding process
Normal Flow	• Doctor accesses patient record • Enters diagnosis findings • Documents procedures performed • Adds clinical notes • Saves encounter details
Alternative Flow	• Voice-to-text documentation - use speech recognition • Template-based entry - select from pre-defined templates • Incomplete documentation - save as draft for later completion
Special Requirement	Medical terminology support, voice recognition capability, secure data storage
Frequency of Use	High
Assumption	Doctor has completed patient examination and has all clinical information

6.5. Submit for Coding

Table 26: EUC-05 Submit for coding

Use case ID	05
Use case Name	Submit for Coding
Actor	Doctor/Clinician
Description	Doctor finalizes encounter documentation and submits it to medical coding department for ICD and CPT code assignment.
Trigger	Doctor completes entering all encounter details
Pre-Condition	Encounter details are fully documented
Post Condition	Encounter is queued for medical coder review and coding
Normal Flow	• Doctor reviews entered details • Clicks submit for coding button • System validates completeness • Encounter is added to coding queue • Notification sent to medical coder
Alternative Flow	• Missing required fields - display validation errors • Urgent case - mark as priority in coding queue • Additional documentation needed - return to doctor with comments
Special Requirement	Workflow notification system, queue management
Frequency of Use	High
Assumption	All required clinical documentation is complete and accurate

6.6. Access Encounter Data

Table 27: EUC-06 Access Encounter Data

Use case ID	06
Use case Name	Access Encounter Data
Actor	Medical Coder
Description	Medical coder retrieves and reviews encounter documentation submitted by doctors to prepare for code assignment.
Trigger	Medical coder starts coding workflow or reviews pending encounters
Pre-Condition	Encounters are submitted for coding, coder is authenticated
Post Condition	Coder has full access to encounter details for coding process
Normal Flow	• Coder logs into system • Views pending encounters queue • Selects encounter to code • Reviews diagnosis and procedure documentation • Accesses patient history if needed
Alternative Flow	• Insufficient documentation - request clarification from doctor • Search specific encounter - use search filters • Priority cases - sort by urgency
Special Requirement	Search functionality, document viewer, secure data access
Frequency of Use	High
Assumption	Encounter documentation is complete and legible

6.7. Assign ICD & CPT Codes

Table 28: EUC-07 Assign ICD & CPT Codes

Use case ID	07
Use case Name	Assign ICD & CPT Codes
Actor	Medical Coder
Description	Medical coder assigns appropriate ICD diagnosis codes and CPT procedure codes based on clinical documentation.
Trigger	Coder accesses encounter data for coding
Pre-Condition	Encounter documentation is reviewed and understood
Post Condition	Accurate ICD and CPT codes are assigned to the encounter
Normal Flow	• Coder analyzes diagnosis documentation • Searches and selects appropriate ICD codes • Reviews procedures performed • Assigns corresponding CPT codes • Verifies code accuracy
Alternative Flow	• Ambiguous documentation - request doctor clarification • ML code recommendations available - review and accept/modify suggestions • Multiple diagnoses - assign all applicable codes
Special Requirement	Updated ICD-10 and CPT code database, code search functionality
Frequency of Use	High
Assumption	Coder has medical coding certification and understands clinical terminology

6.8. Generate Code Recommendations

Table 29: EUC-08 Generate Code Recommendations

Use case ID	08
Use case Name	Attend Live Session
Actor	System (ML/AI Module)
Description	AI system analyzes encounter documentation and suggests relevant ICD and CPT codes to assist medical coder.
Trigger	Coder opens encounter for code assignment
Pre-Condition	ML model is trained, encounter text is available
Post Condition	System displays recommended codes with confidence scores
Normal Flow	• System extracts text from encounter • NLP processes clinical terms • ML model predicts relevant codes • Displays recommendations to coder • Coder reviews and accepts/modifies
Alternative Flow	• Low confidence scores - coder performs manual search • No recommendations - proceed with manual coding • Feedback on recommendations - improve ML model
Special Requirement	Trained ML model, NLP engine, real-time processing capability
Frequency of Use	Normal
Assumption	ML model is regularly updated with latest coding patterns

6.9. Submit Coded Data

Table 30: EUC-09 Submit Coded Data

Use case ID	09
Use case Name	Submit Coded Data
Actor	Medical Coder
Description	Medical coder finalizes code assignment and submits coded encounter to billing department for claim generation.
Trigger	Coder completes code assignment and verification
Pre-Condition	All required codes are assigned and validated
Post Condition	Coded encounter is available to billing staff for claim creation
Normal Flow	• Coder reviews all assigned codes • Verifies code accuracy and completeness • Adds coding notes if needed • Submits to billing queue • System notifies billing staff
Alternative Flow	• Codes require review - flag for supervisor review • Incomplete coding - save draft and complete later • Urgent billing - mark as priority
Special Requirement	Quality assurance workflow, audit trail logging
Frequency of Use	Normal
Assumption	All codes comply with current coding guidelines and payer requirements

6.10. Generate Claim

Table 31: EUC-10 Generate Claim

Use case ID	10
Use case Name	Generate Claim
Actor	Billing Staff
Description	Billing staff creates insurance claim using coded encounter data, patient information, and insurance details.
Trigger	Coded encounter is received in billing queue
Pre-Condition	Encounter is coded, patient insurance information is verified
Post Condition	Insurance claim is generated and ready for validation
Normal Flow	• Billing staff accesses coded encounter • System populates claim form with patient and coding data • Staff verifies insurance coverage • Adds billing amounts • Generates claim document
Alternative Flow	• Insurance information missing - contact patient for details • Multiple payers - create separate claims for each • Pre-authorization required - verify before claim generation
Special Requirement	CMS-1500/UB-04 form generation, insurance database integration
Frequency of Use	Normal
Assumption	Patient insurance is active and covers the services provided

6.11. Validate Claim

Table 32: EUC-11 Validate Claim

Use case ID	11
Use case Name	Validate Claim
Actor	Billing Staff
Description	Billing staff reviews generated claim for accuracy, completeness, and compliance with payer requirements before submission.
Trigger	Claim is generated and ready for review
Pre-Condition	Claim is created with all required fields populated
Post Condition	Claim is validated and approved for submission to insurance payer
Normal Flow	• Staff reviews claim details • Verifies code-diagnosis compatibility • Checks payer-specific requirements • Validates billing amounts • Approves claim for submission
Alternative Flow	• Errors found - correct and regenerate claim • AI prediction shows high rejection risk - review and modify • Missing information - request from appropriate department
Special Requirement	Claim validation rules engine, payer requirement database
Frequency of Use	Normal
Assumption	Staff is familiar with insurance payer requirements and claim standards

6.12. Predict Rejection

Table 33: EUC-12 Predict Rejection

Use case ID	12
Use case Name	Predict Rejection
Actor	System (ML/AI Module)
Description	AI system analyzes claim data against historical rejection patterns and predicts likelihood of claim rejection.
Trigger	Claim validation process is initiated
Pre-Condition	Claim data is complete, ML model is trained on historical claims
Post Condition	Rejection probability score and reasons are displayed to billing staff
Normal Flow	• System extracts claim features • ML model analyzes against rejection patterns • Calculates rejection probability • Identifies potential rejection reasons • Displays risk score to staff
Alternative Flow	• High rejection risk - highlight specific issues for correction • Model uncertainty - proceed with manual review • New payer with limited history - flag for careful review
Special Requirement	Historical claims database, trained ML model, real-time prediction
Frequency of Use	High
Assumption	ML model is trained on sufficient historical claim and rejection data

6.13. Submit Claim

Table 34: EUC-13 Submit Claim

Use case ID	13
Use case Name	Submit Claim
Actor	Billing Staff
Description	Billing staff electronically submits validated claim to insurance payer through clearinghouse or direct submission.
Trigger	Claim is validated and approved for submission
Pre-Condition	Claim passes all validation checks, payer information is correct
Post Condition	Claim is submitted and tracking number is received
Normal Flow	• Staff selects validated claim • Clicks submit button • System formats claim to payer specifications • Transmits claim electronically • Receives submission confirmation
Alternative Flow	• Submission fails - retry or submit via alternate method • Batch submission - submit multiple claims simultaneously • Paper claim required - print and mail physical form
Special Requirement	Availability of Internet
Frequency of Use	High
Assumption	Electronic submission infrastructure is operational and payer accepts electronic claims

6.14. Track Claim Status

Table 35: EUC-14 Track Claim Status

Use case ID	14
Use case Name	Track Claim Status
Actor	Insurance Payer, Billing Staff
Description	System monitors claim status with insurance payer and updates billing staff on acceptance, rejection, or payment processing.
Trigger	Claim is submitted or billing staff requests status update
Pre-Condition	Claim is submitted to payer, tracking number exists
Post Condition	Current claim status is displayed and logged in system
Normal Flow	• System queries payer for claim status • Receives status update (pending/accepted/rejected/paid) • Updates claim record• Notifies billing staff of status change • Displays current status
Alternative Flow	• Claim rejected - route to rejection handling workflow • Additional information requested - notify appropriate staff • Payment received - proceed to payment processing
Special Requirement	Payer API integration, automated status polling, notification system
Frequency of Use	High
Assumption	Insurance payer provides electronic status inquiry capability

6.15. View Billing Status

Table 36: EUC-15 View Billing Status

Use case ID	15
Use case Name	View Billing Status
Actor	Patient
Description	Patient accesses their billing information to view charges, insurance payments, outstanding balance, and claim status.
Trigger	Patient wants to check billing or claim status
Pre-Condition	Patient is logged into the system, billing records exist
Post Condition	Patient views current billing status and claim progress
Normal Flow	• Patient logs in • Navigates to billing section • Views list of encounters and charges • Checks insurance claim status • Reviews payment history and balance
Alternative Flow	• Billing questions - access help or contact billing staff • Download statements - export to PDF • Set up payment plan - access payment options
Special Requirement	Patient portal, real-time data access, user-friendly interface
Frequency of Use	High
Assumption	Patient has internet access and basic computer literacy

6.16. Manage Users

Table 37: EUC-16 Manage Users

Use case ID	16
Use case Name	Manage Users
Actor	System Administrator
Description	Administrator creates, modifies, deactivates user accounts and assigns appropriate roles and permissions within the system.
Trigger	New user needs access or existing user requires permission changes
Pre-Condition	Administrator is authenticated with admin privileges
Post Condition	User account is created/modified and user has appropriate system access
Normal Flow	• Admin accesses user management interface • Creates new user or selects existing user • Assigns role (doctor, coder, billing staff, patient) • Sets permissions and access levels • Saves user configuration
Alternative Flow	• Deactivate user - disable account for terminated employees • Reset password - initiate password reset for user • Bulk user creation - import multiple users from file
Special Requirement	Role-based access control, audit logging, secure password policies
Frequency of Use	Low to Medium
Assumption	Administrator understands organizational roles and appropriate permission levels

6.17. Configure ML Models

Table 38: EUC-17 Configure ML Models

Use case ID	17
Use case Name	Configure ML Models
Actor	System
Description	Administrator configures AI/ML model parameters, retrains models with new data, and monitors model performance metrics.
Trigger	Model performance degrades or new training data is available
Pre-Condition	Admin has ML configuration access, training data is prepared
Post Condition	ML model is updated, performance metrics are logged, model is deployed
Normal Flow	• Admin accesses ML configuration dashboard • Reviews current model performance • Selects training data set • Configures model parameters • Initiates training process • Validates and deploys model
Alternative Flow	•Model performance inadequate - adjust hyper parameters and retrain • Emergency rollback - revert to previous model version • A/B testing - deploy models for comparison
Special Requirement	ML training infrastructure, model versioning, performance monitoring
Frequency of Use	High
Assumption	Administrator has ML/AI technical knowledge and understands model evaluation metrics

7. Domain Model

The Domain Model Diagram below represents the conceptual data structure of the HealPay Medical Billing System by identifying key entities and their relationships. The diagram illustrates core business entities including Patient, Doctor, Medical Coder, Medical Records, ICD/CPT Codes, Claims, Invoices, Payments, and Insurance Providers, along with their attributes and associations. It models the static structure of the system, showing cardinality relationships (one-to-one, one-to-many, many-to-many) that define how entities interact and constrain the business logic for patient encounters, medical coding, claim processing, and payment transactions.

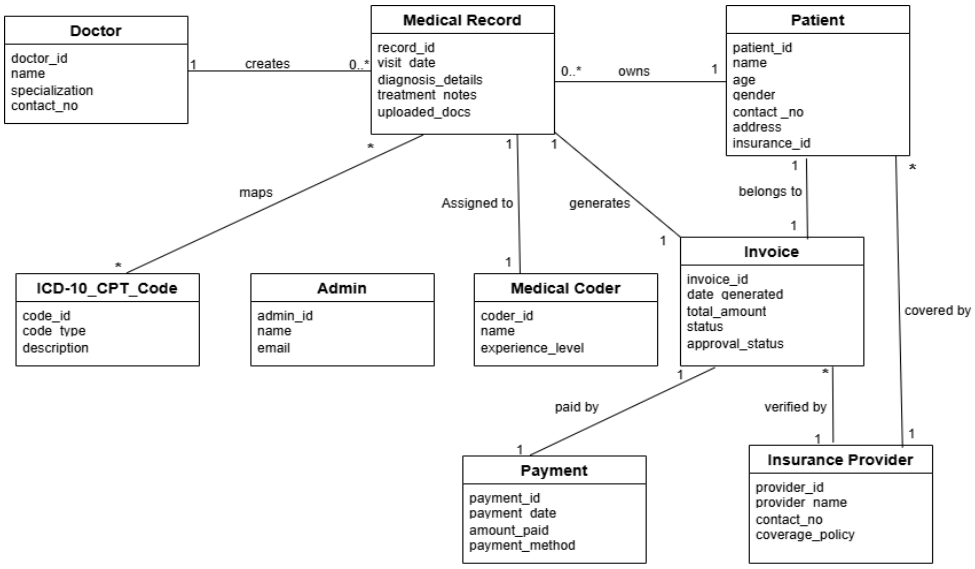

Figure 2: Domain Model For HealPay

8. ERD Diagram

The Entity Relationship Diagram below defines the logical database structure of the HealPay Medical Billing System, showing entities, attributes, primary keys, foreign keys, and relationship cardinalities. The diagram models data entities including Patient, Doctor, Medical Coder, Medical Records, ICD/CPT Codes, Claims, Invoices, Payments, and Insurance Providers, with their complete attribute definitions and referential integrity constraints. It specifies relationship cardinalities (1:1, 1:M, M:N) through notations that define data dependencies and ensure data consistency across patient encounters, medical coding, claim generation, and payment processing workflows.

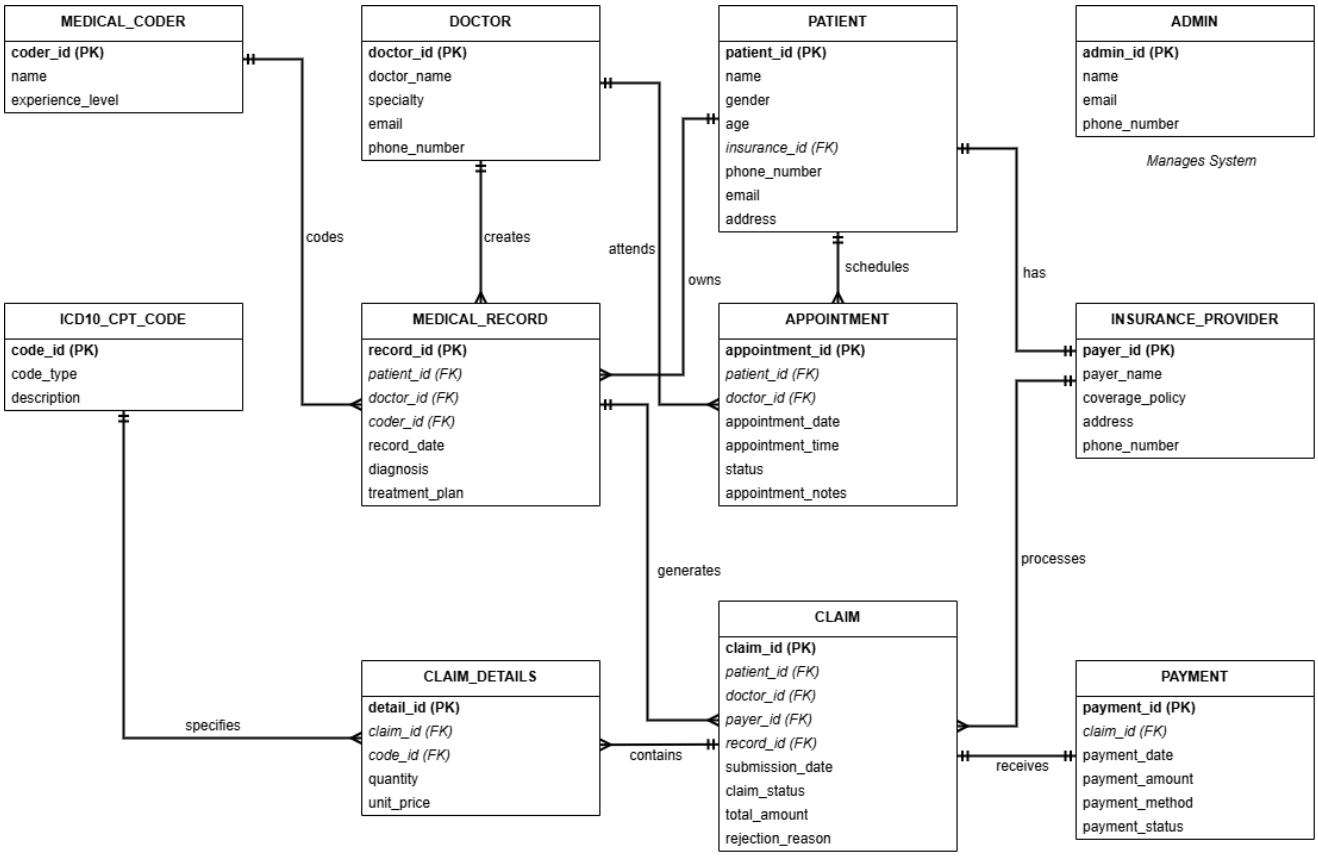

Figure 3: ERD Diagram For HealPay

9. Activity Diagram

This Activity Diagram models the dynamic workflow of the HealPay Medical Billing System by representing sequential and parallel activities across different actors using swimlanes for Patient, Doctor/Clinician, Billing Staff, and Automated System processes. The diagram illustrates complete business processes from patient intake through OCR processing, medical coding, claim generation, validation, submission to insurance payers, payment processing, and anomaly detection. It includes decision points, merge points, and parallel processing activities, showing how different stakeholders collaborate throughout the billing lifecycle while automated systems perform background operations such as text extraction, code recommendations, and payment reconciliation.

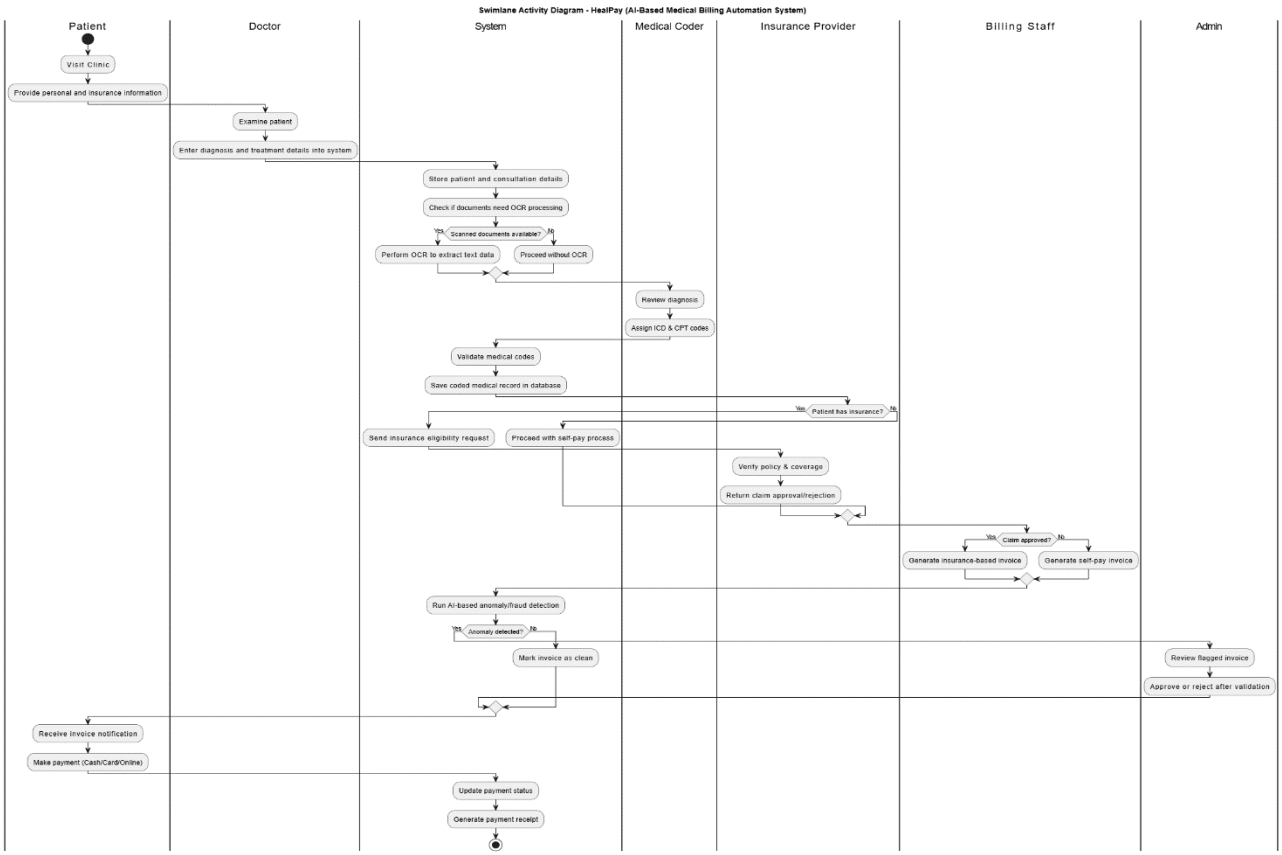

Figure 4: Activity Diagram for HealPay

10. Package Diagram

This Package Diagram represents the logical organization of the HealPay Medical Billing System by grouping related classes and components into packages that reflect the system's modular architecture. The diagram shows package hierarchies including UI Packages (Patient Interface, Provider Interface, Admin Interface), Backend Packages (Authentication, Business Logic, Data Access), and Integration Packages (OCR/NLP, ML/AI, Payment Gateway, Insurance Integration). It illustrates package relationships through import, access, and uses dependencies, demonstrating separation of concerns, modular design principles, and how packages interact to deliver comprehensive medical billing functionality while maintaining code organization and reusability.

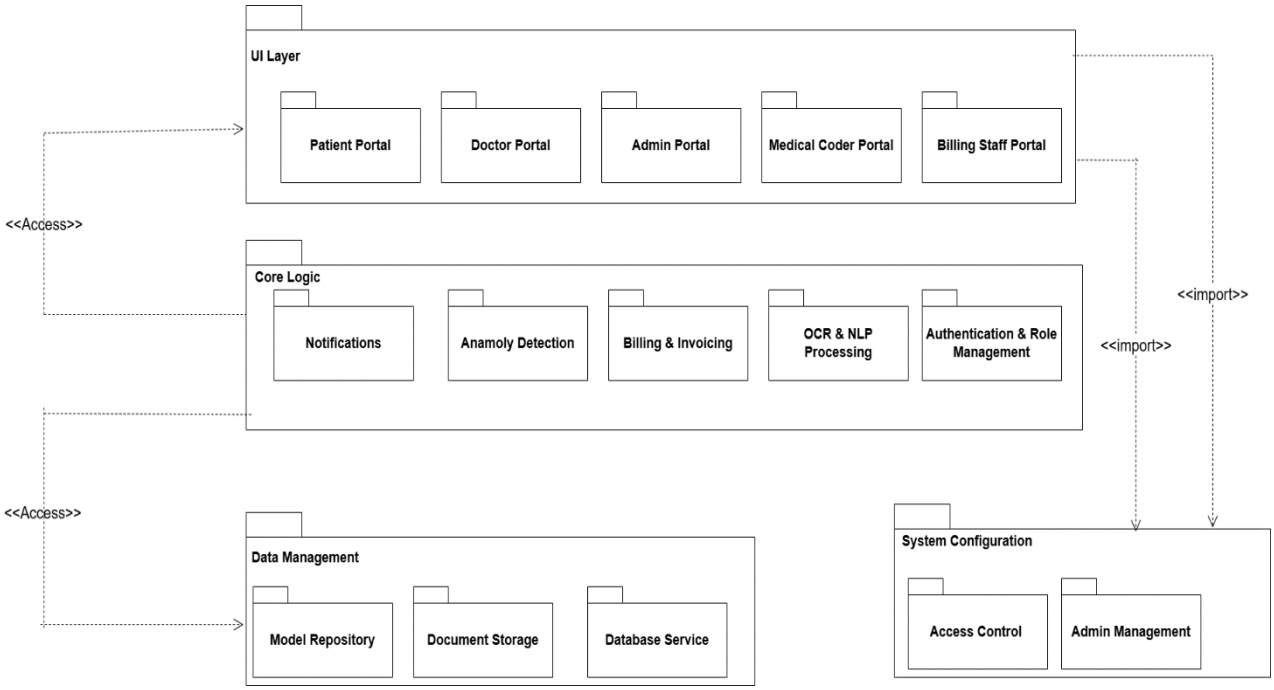

Figure 5: Package Diagram For HealPay

11. Component Diagram

This Component Diagram illustrates the physical architecture of the HealPay Medical Billing System by depicting major software components, their interfaces, and dependencies across different architectural layers. The diagram organizes components into logical layers including User Interface (Patient Portal, Doctor Portal, Billing Portal), Application Services (Authentication, OCR/NLP Module, Code Mapping, Billing Engine), Data Access Layer, and external integrations. It shows component interfaces (provided/required), port connections, and communication patterns, demonstrating how components collaborate through defined interfaces to support medical billing operations, AI/ML processing, and system scalability.

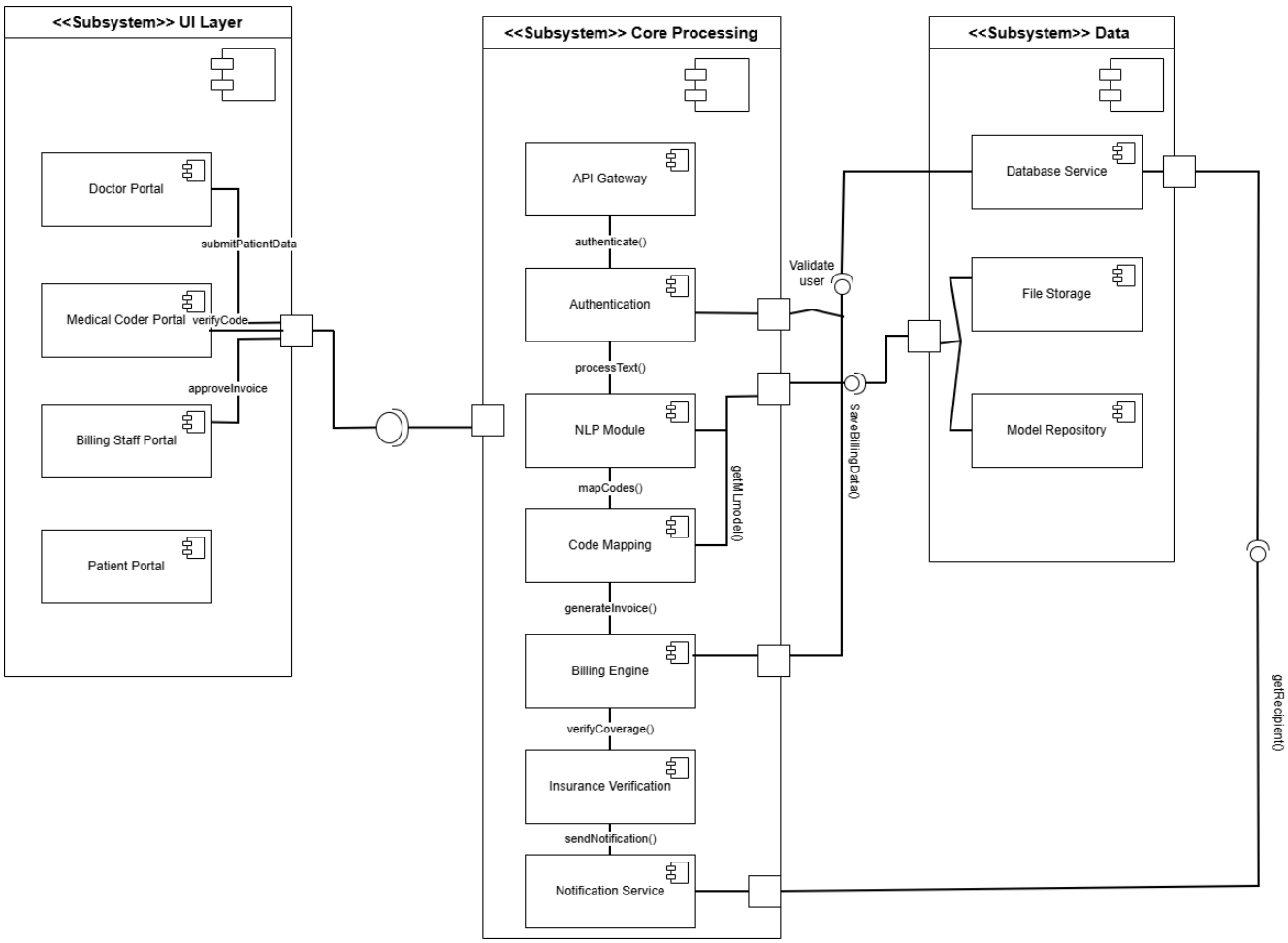

Figure 6: Component Diagram for HealPay

12. Data Flow Diagrams

12.1. Data Flow Diagram Level 0

This Level 0 Data Flow Diagram provides the highest-level view of the HealPay Medical Billing System by depicting the entire system as a single process node connected to external entities. The diagram shows data flows between the system and six external entities (Patient, Doctor, Medical Coder, Billing Staff, Insurance Provider, and Admin), illustrating what data enters and exits the system at its boundaries. It establishes the system's scope and context, identifying major input and output data streams that represent the complete medical billing process from patient information intake to payment processing and administrative management.

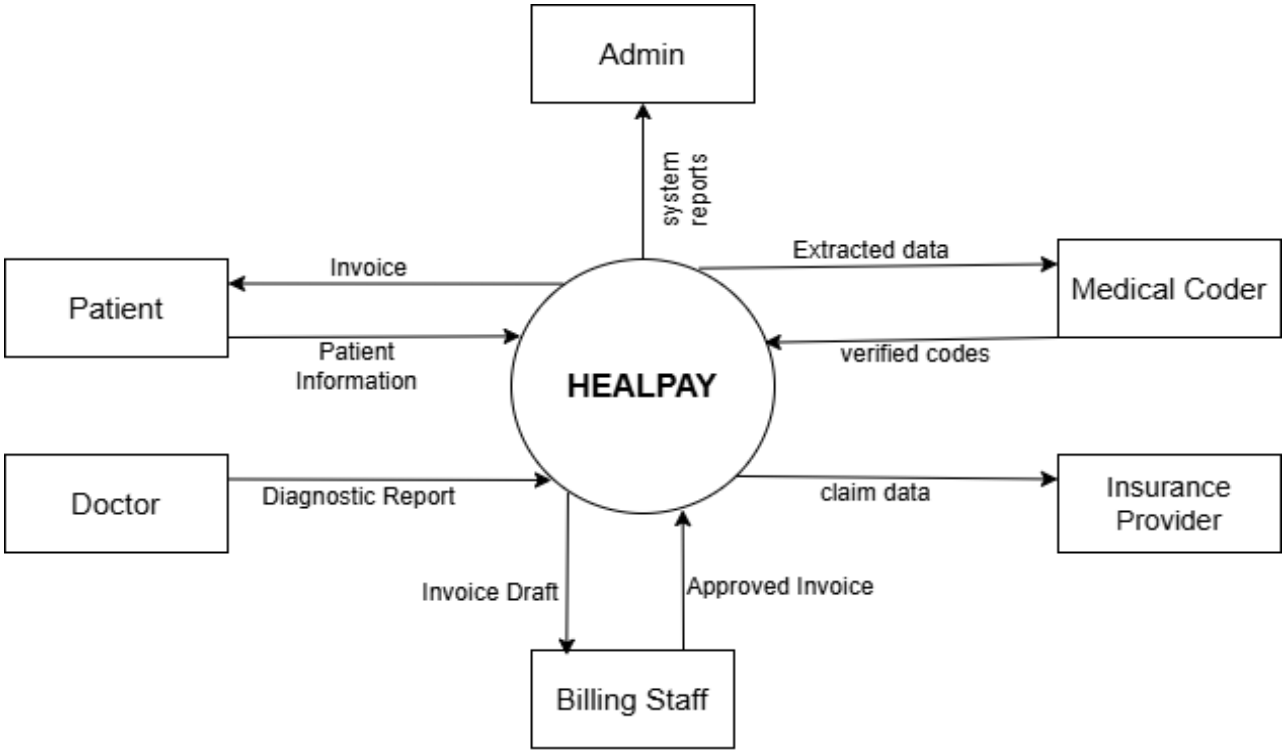

Figure 7 : DFD level 0 for HealPay

12.2. Data Flow Diagram Level 1

This Level 1 Data Flow Diagram decomposes the HealPay Medical Billing System into major high-level processes representing core functional areas of the system. The diagram shows six primary processes (Patient Data Intake, Code Mapping, Billing & Invoice Generation, Insurance Verification, Payment Processing, and Anomaly Detection) with data flows between them and to external entities. It illustrates how data moves through data stores (Patient Records, Medical Records, ICD/CPT Coding, Invoices, Payments, Audit Log) and demonstrates the sequence of operations from data collection through coding, billing, claim submission, and payment handling, providing a clear functional overview of the system's data transformations.

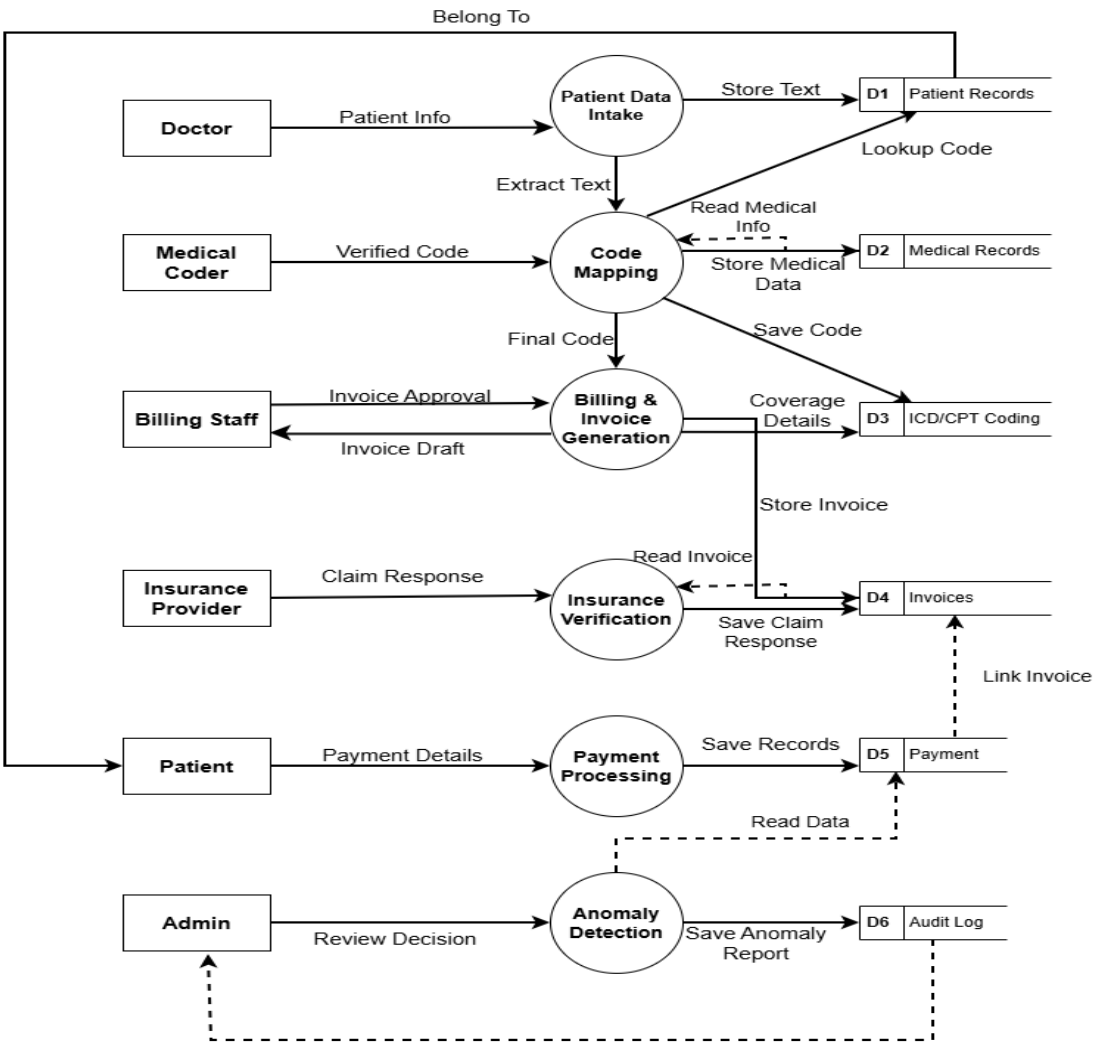

Figure 8 : DFD level 1 for HealPay

12.3. Data Flow Diagram Level 2

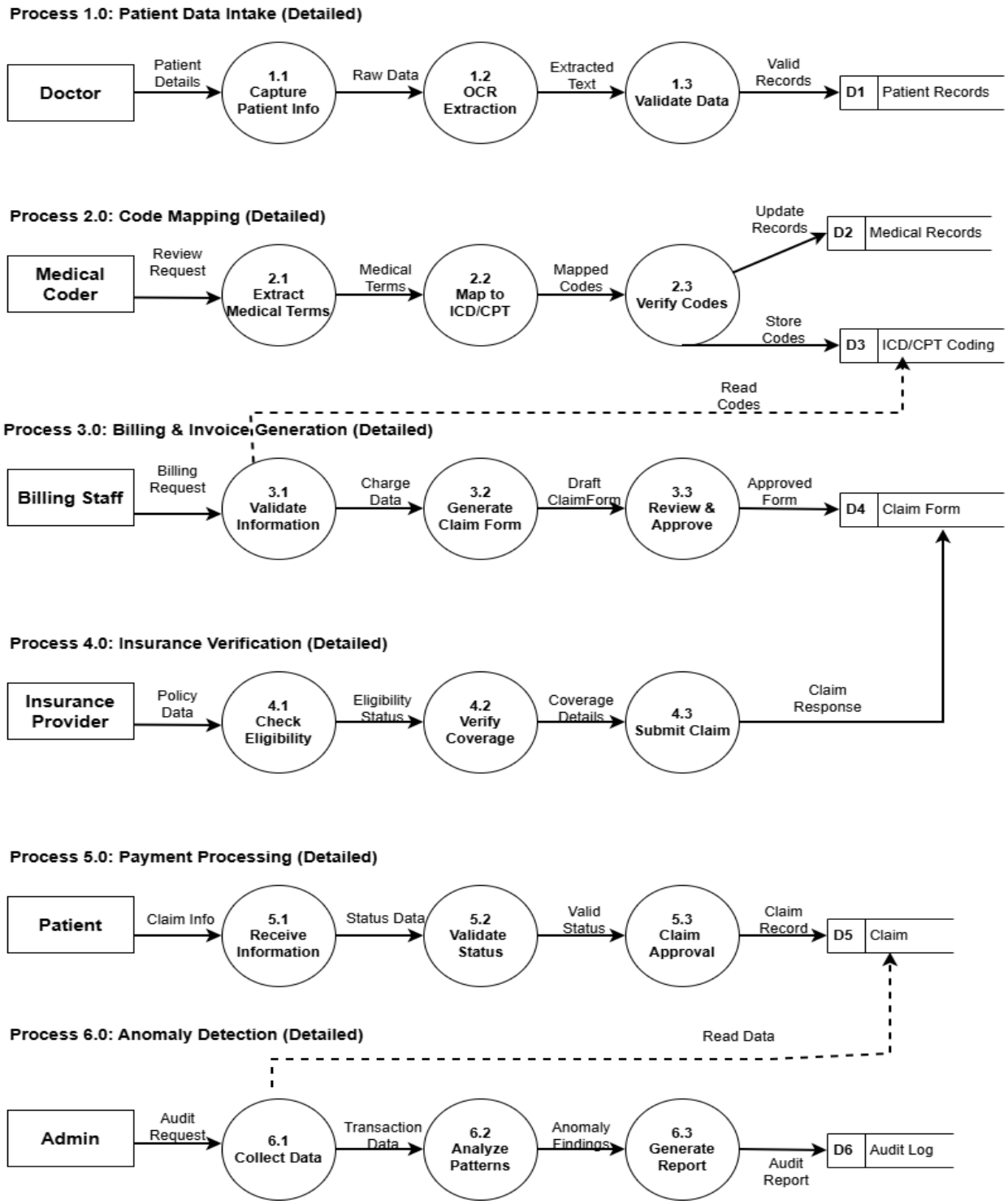

Figure 9 : DFD level 2 for HealPay

13. System Sequence Diagram

13.1. Register/Login

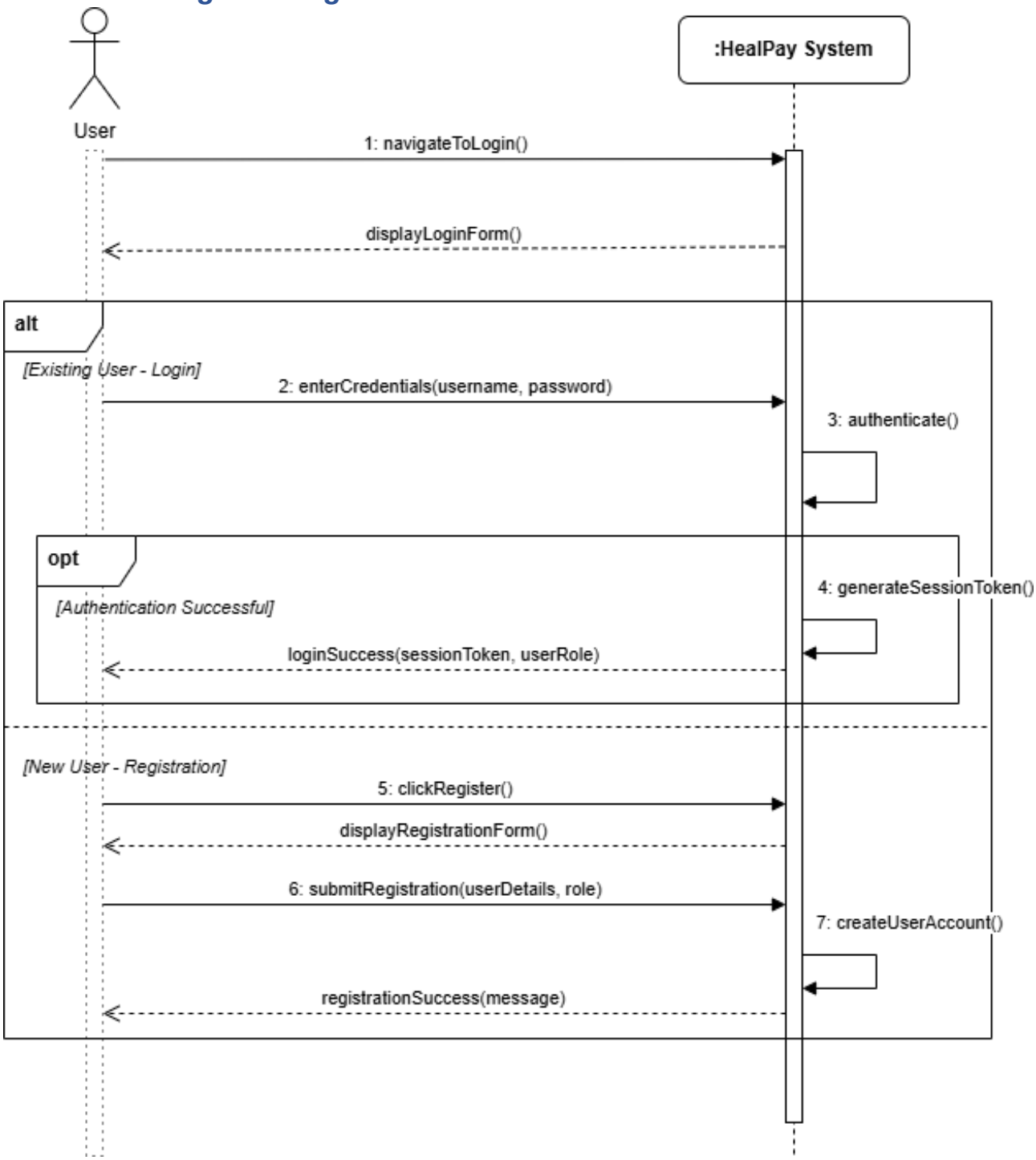

Figure 10 : Register/Login SSD for HealPay

13.2. Complete Intake Form

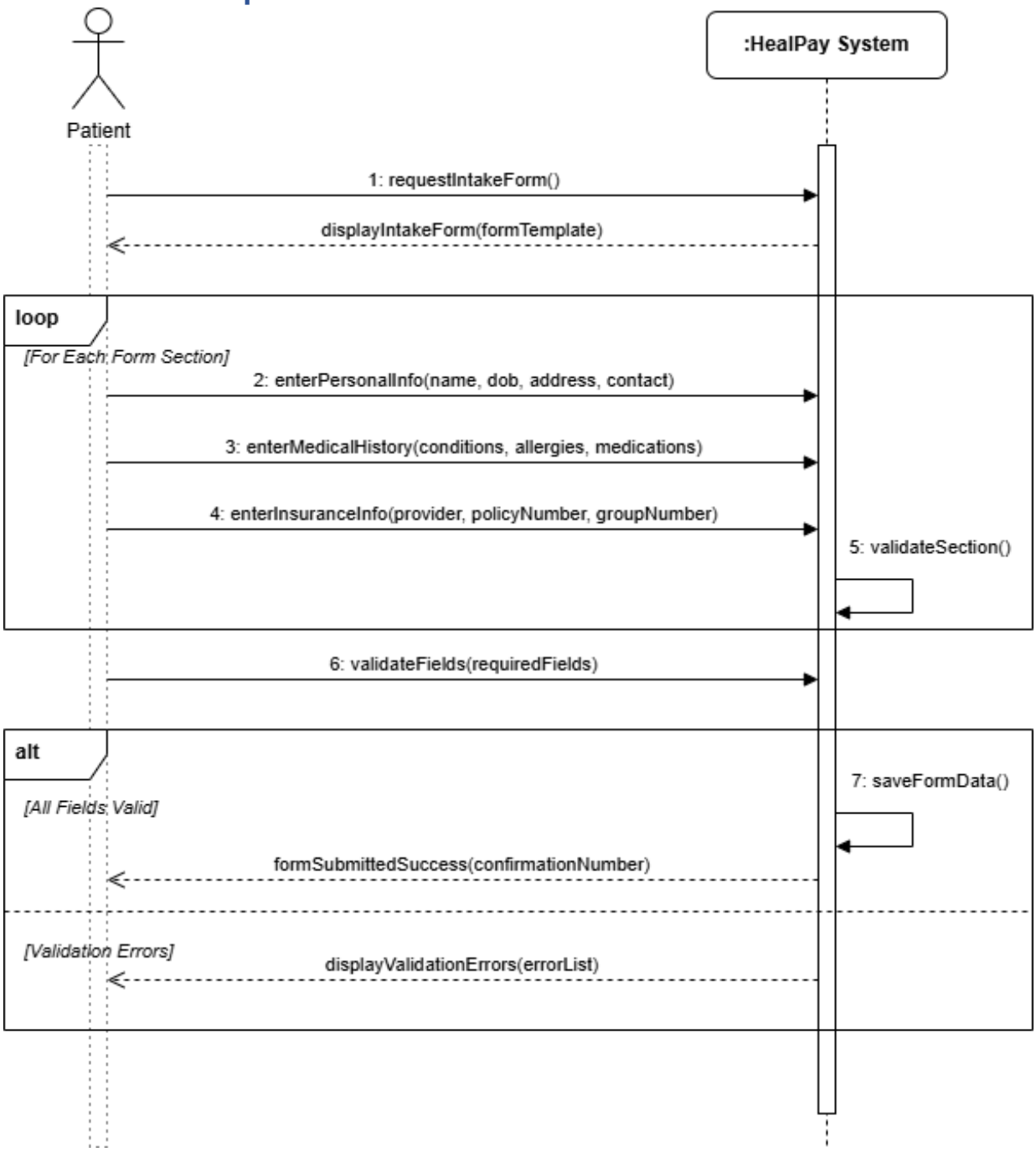

Figure 11 : Complete Intake Form SSD for HealPay

13.3. Assign Code

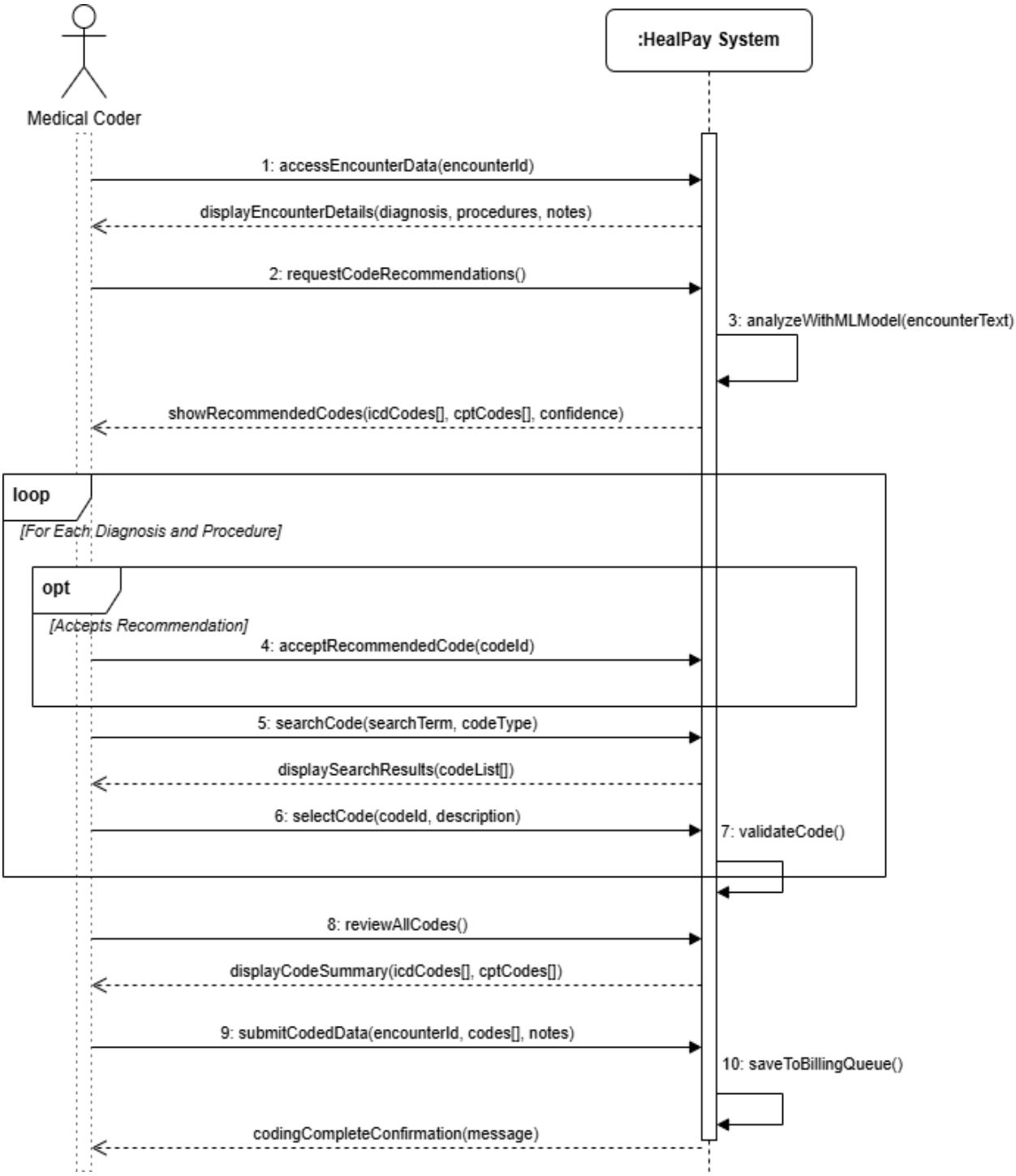

Figure 12 : Assign code SSD for HealPay

13.4. Generate & Validate Claim

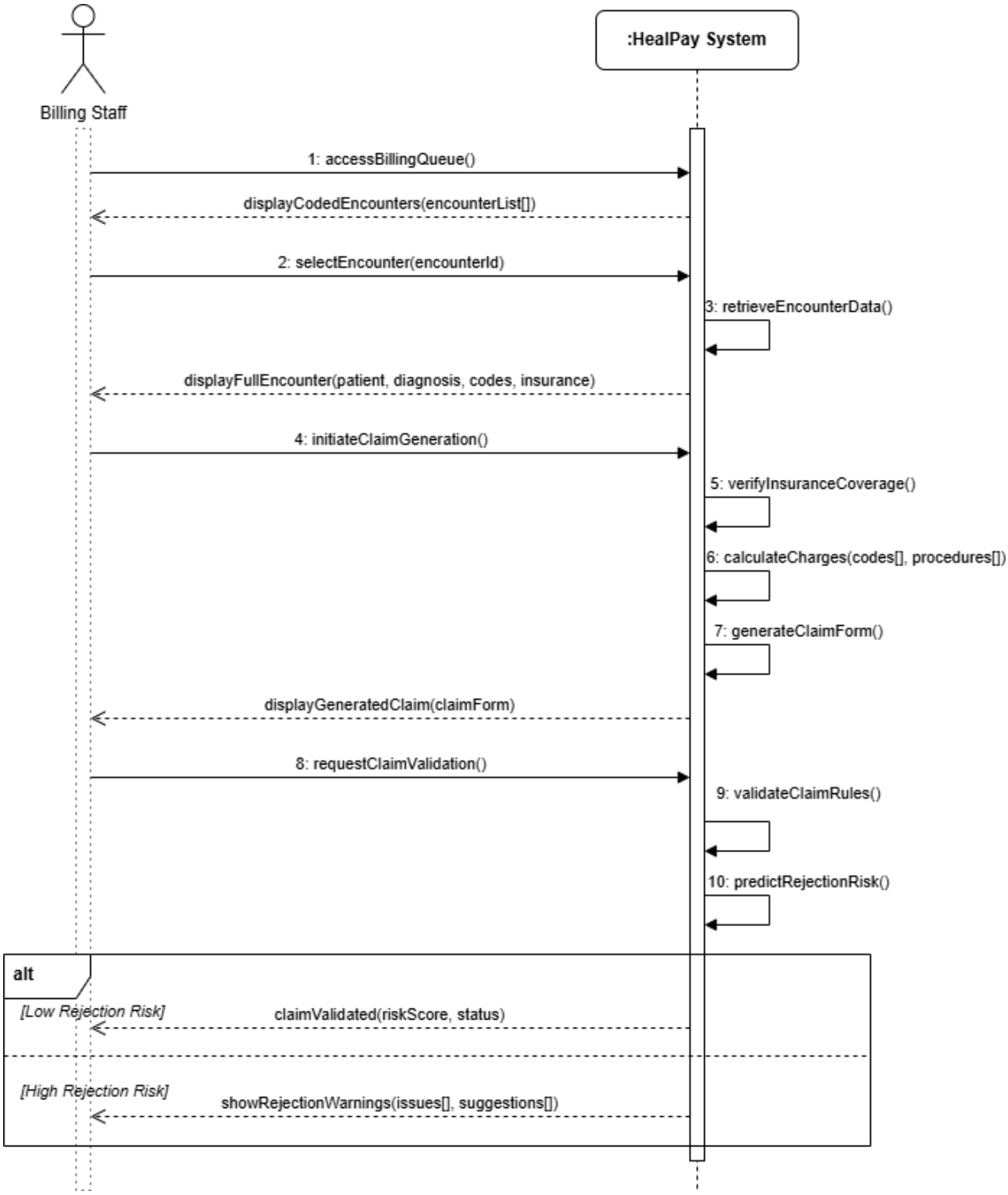

Figure 13 : Generate claim SSD for HealPay

13.5. Submit and Track Claim Status

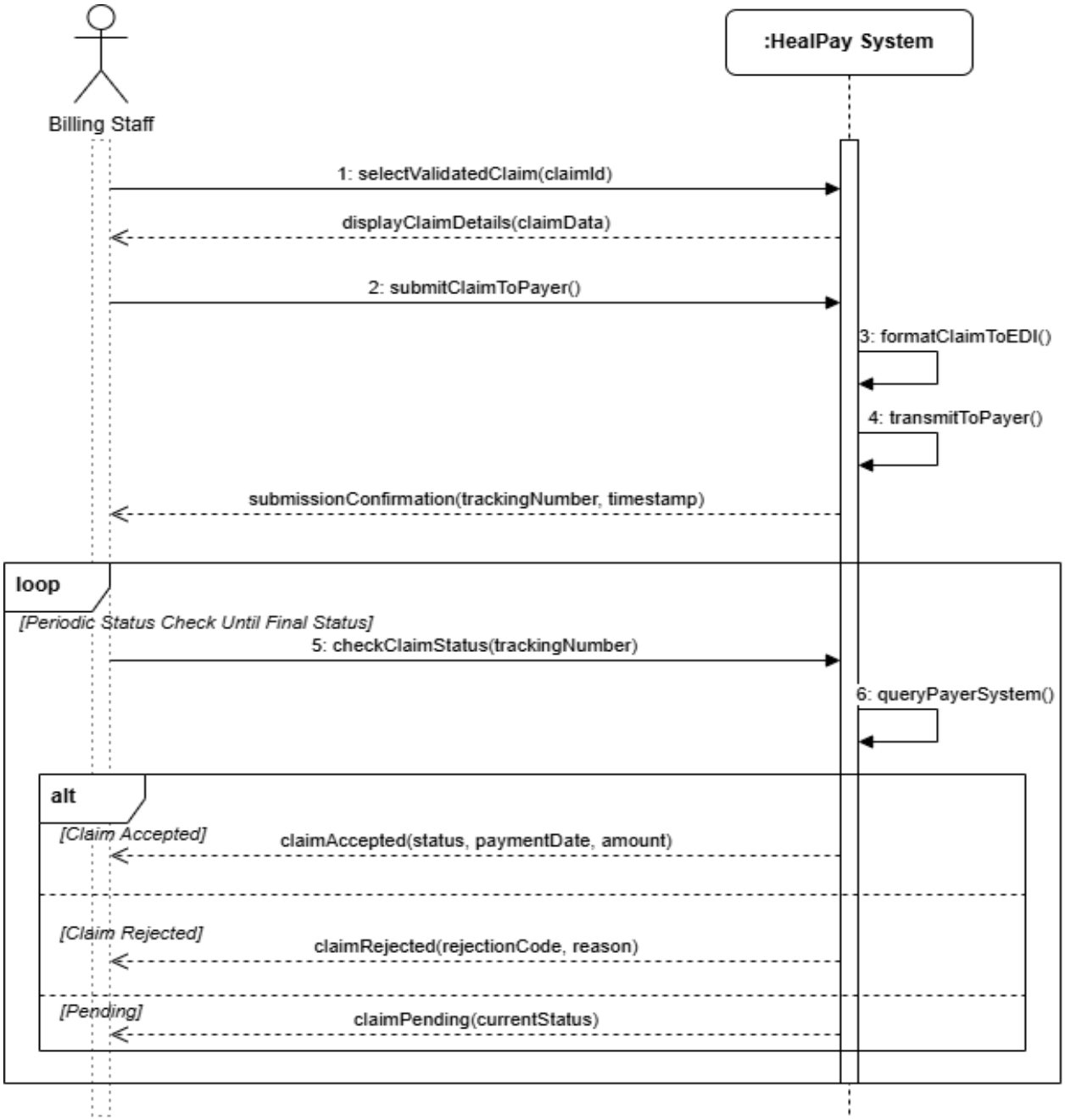

Figure 14: Claim status SSD for HealPay

13.6. Manage Users

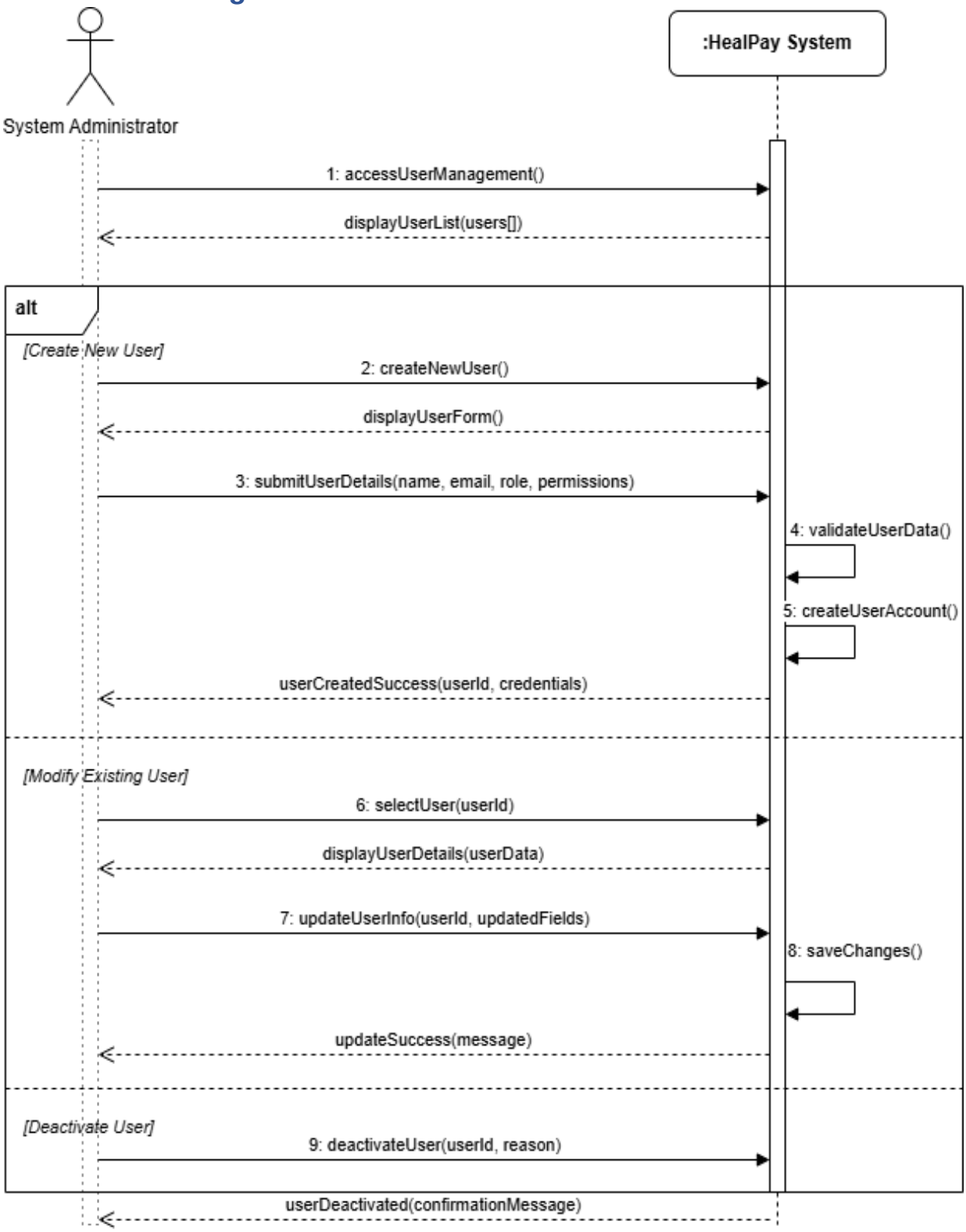

Figure 15:Manage Users SSD for HealPay

14. High Level Architecture Diagram

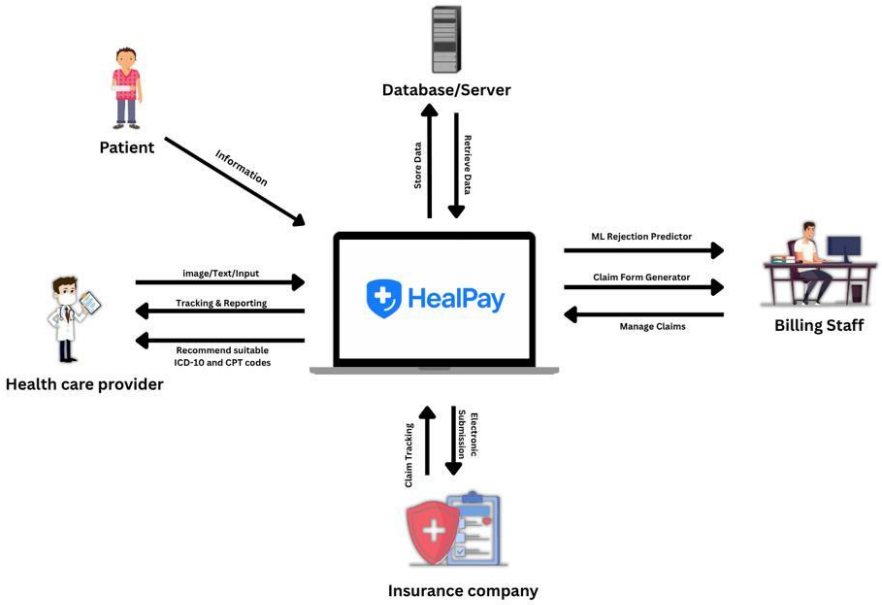

Figure 16: High Level Architecture For HealPay

15. References

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- https://www.kaggle.com/datasets/prasad22/healthcare-dataset
- https://apps.diagrams.net